

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
df = pd.read_csv("n.csv")
df.shape
df.dtypes
df.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	descript
0	s10000	Movie	Netflix Original 0	Christopher Nolan	Leonardo DiCaprio, Kate Winslet	South Korea	January 01, 2015	2001	R	87 min	Children & Family Movies	suspens mystery keeps guess
1	s10001	TV Show	Netflix Original 1	Ava DuVernay	NaN	United Kingdom	January 06, 2015	1999	NaN	129 min	Documentaries	A fast-pa ac adventure of twi
2	s10002	Movie	Netflix Original 2	Greta Gerwig	Emma Stone, Ryan Gosling	United States	January 11, 2015	1995	PG-13	100 min	Dramas, International Movies	A chil horror s that haunt y
3	s10003	Movie	Netflix Original 3	Ava DuVernay	Chris Evans, Scarlett Johansson	United States	January 16, 2015	2007	PG-13	75 min	Dramas, International Movies	A roma tale spans acr decac
4	s10004	Movie	Netflix Original 4	Christopher Nolan	Leonardo DiCaprio, Kate Winslet	NaN	January 21, 2015	2015	NaN	NaN	Children & Family Movies	A fast-pa ac adventure of twi

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
Drop=df.drop(columns=['cast', 'director', 'description'])
print("Dropped Columns:\n",Drop)
df=Drop
```

Dropped Columns:

	show_id	type	title	country	date_added	\
0	s10000	Movie	Netflix Original 0	South Korea	January 01, 2015	
1	s10001	TV Show	Netflix Original 1	United Kingdom	January 06, 2015	
2	s10002	Movie	Netflix Original 2	United States	January 11, 2015	
3	s10003	Movie	Netflix Original 3	United States	January 16, 2015	
4	s10004	Movie	Netflix Original 4	NaN	January 21, 2015	
..	...	...	...	...	...	
495	s10495	Movie	Netflix Original 495	United Kingdom	October 11, 2021	
496	s10496	Movie	Netflix Original 496	NaN	October 16, 2021	
497	s10497	TV Show	Netflix Original 497	India	October 21, 2021	
498	s10498	Movie	Netflix Original 498	South Korea	October 26, 2021	
499	s10499	NaN	Netflix Original 499	United States	October 31, 2021	

	release_year	rating	duration	listed_in
0	2001	R	87 min	Children & Family Movies
1	1999	NaN	129 min	Documentaries
2	1995	PG-13	100 min	Dramas, International Movies
3	2007	PG-13	75 min	Dramas, International Movies
4	2015	NaN	NaN	Children & Family Movies
..	...	...	...	...
495	2013	PG	80 min	Documentaries
496	2013	TV-PG	81 min	Children & Family Movies
497	2019	R	4 Seasons	Comedies, Dramas
498	2015	TV-MA	89 min	Horror Movies
499	2006	TV-PG	NaN	Children & Family Movies

[500 rows x 9 columns]

```
df=df.dropna()
df
```

	show_id	type	title	country	date_added	release_year	rating	duration	listed_in
0	s10000	Movie	Netflix Original 0	South Korea	January 01, 2015	2001	R	87 min	Children & Family Movies
2	s10002	Movie	Netflix Original 2	United States	January 11, 2015	1995	PG-13	100 min	Dramas, International Movies
3	s10003	Movie	Netflix Original 3	United States	January 16, 2015	2007	PG-13	75 min	Dramas, International Movies
6	s10006	Movie	Netflix Original 6	India	January 31, 2015	2012	PG	4 Seasons	Romantic Movies
7	s10007	Movie	Netflix Original 7	United Kingdom	February 05, 2015	2007	TV-14	4 Seasons	Dramas, International Movies
...	...	...	...	...	...	...	...	...	...
490	s10490	TV Show	Netflix Original 490	South Korea	September 16, 2021	2017	PG-13	138 min	Horror Movies
491	s10491	Movie	Netflix Original 491	Japan	September 21, 2021	2020	PG-13	3 Seasons	Children & Family Movies
495	s10495	Movie	Netflix Original 495	United Kingdom	October 11, 2021	2013	PG	80 min	Documentaries
497	s10497	TV Show	Netflix Original 497	India	October 21, 2021	2019	R	4 Seasons	Comedies, Dramas
498	s10498	Movie	Netflix Original 498	South Korea	October 26, 2021	2015	TV-MA	89 min	Horror Movies

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```
df=df.drop_duplicates()
df
```

	show_id	type	title	country	date_added	release_year	rating	duration	listed_in
0	s10000	Movie	Netflix Original 0	South Korea	January 01, 2015	2001	R	87 min	Children & Family Movies
2	s10002	Movie	Netflix Original 2	United States	January 11, 2015	1995	PG-13	100 min	Dramas, International Movies
3	s10003	Movie	Netflix Original 3	United States	January 16, 2015	2007	PG-13	75 min	Dramas, International Movies
6	s10006	Movie	Netflix Original 6	India	January 31, 2015	2012	PG	4 Seasons	Romantic Movies
7	s10007	Movie	Netflix Original 7	United Kingdom	February 05, 2015	2007	TV-14	4 Seasons	Dramas, International Movies
...	...	...	...	...	...	...	...	...	...
490	s10490	TV Show	Netflix Original 490	South Korea	September 16, 2021	2017	PG-13	138 min	Horror Movies
491	s10491	Movie	Netflix Original 491	Japan	September 21, 2021	2020	PG-13	3 Seasons	Children & Family Movies
495	s10495	Movie	Netflix Original 495	United Kingdom	October 11, 2021	2013	PG	80 min	Documentaries
497	s10497	TV Show	Netflix Original 497	India	October 21, 2021	2019	R	4 Seasons	Comedies, Dramas
498	s10498	Movie	Netflix Original 498	South Korea	October 26, 2021	2015	TV-MA	89 min	Horror Movies

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df['release_year'].astype(int)
```

```
release_year
```

0	2001
2	1995
3	2007
6	2012
7	2007
...	...
490	2017
491	2020
495	2013
497	2019
498	2015

269 rows × 1 columns

dtype: int64

```
main_genre = df['show_id']  
print("Extracted Column:\n",main_genre)
```

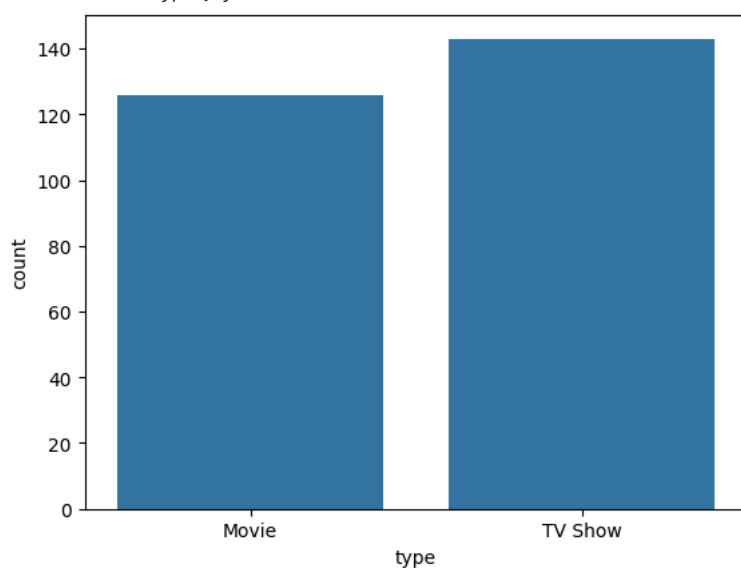
```
Extracted Column:  
0      s10000  
2      s10002  
3      s10003  
6      s10006  
7      s10007  
...  
490    s10490  
491    s10491  
495    s10495  
497    s10497  
498    s10498  
Name: show_id, Length: 269, dtype: object
```

```
df.shape
```

```
(269, 9)
```

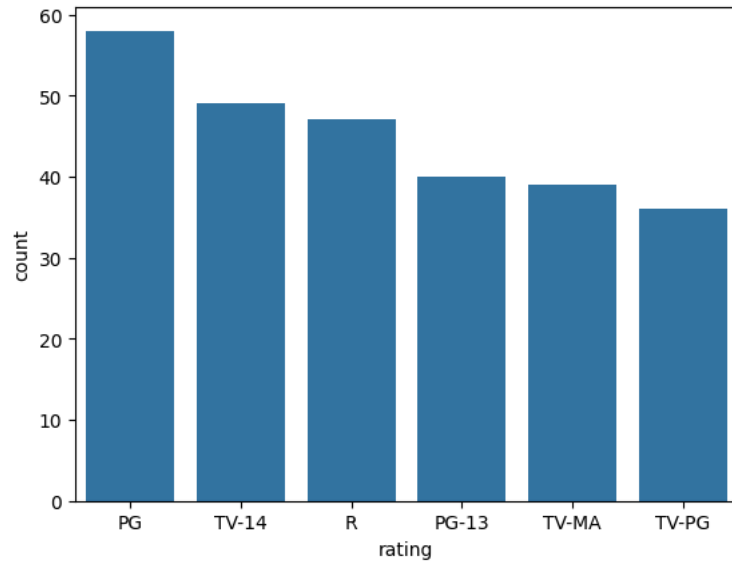
```
import seaborn as sns  
sns.countplot(x='type', data=df)
```

<Axes: xlabel='type', ylabel='count'>



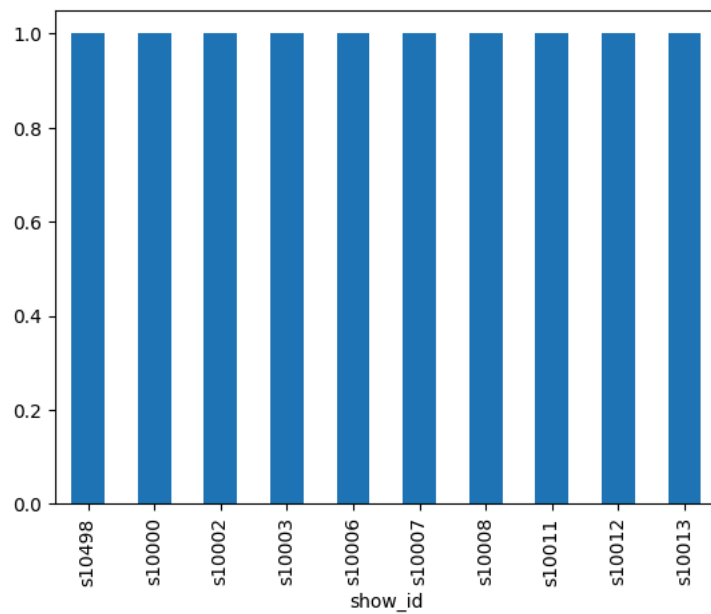
```
sns.countplot(x='rating', data=df, order=df['rating'].value_counts().index)
```

```
<Axes: xlabel='rating', ylabel='count'>
```



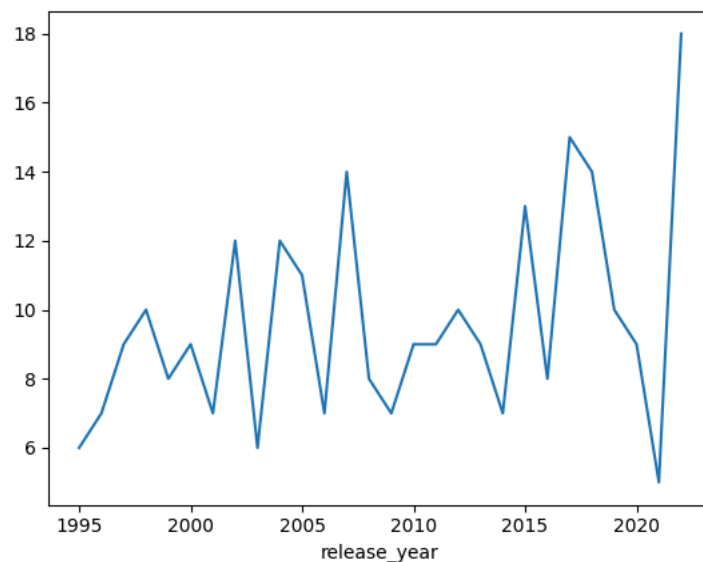
```
main_genre.value_counts().head(10).plot(kind='bar')
```

```
<Axes: xlabel='show_id'>
```



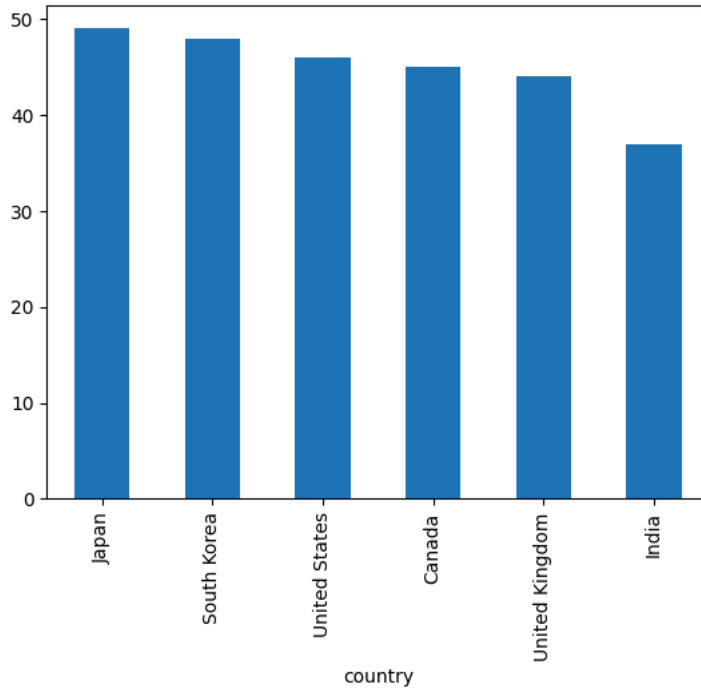
```
df['release_year'].value_counts().sort_index().plot(kind='line')
```

```
<Axes: xlabel='release_year'>
```



```
df['country'].value_counts().head(10).plot(kind='bar')
```

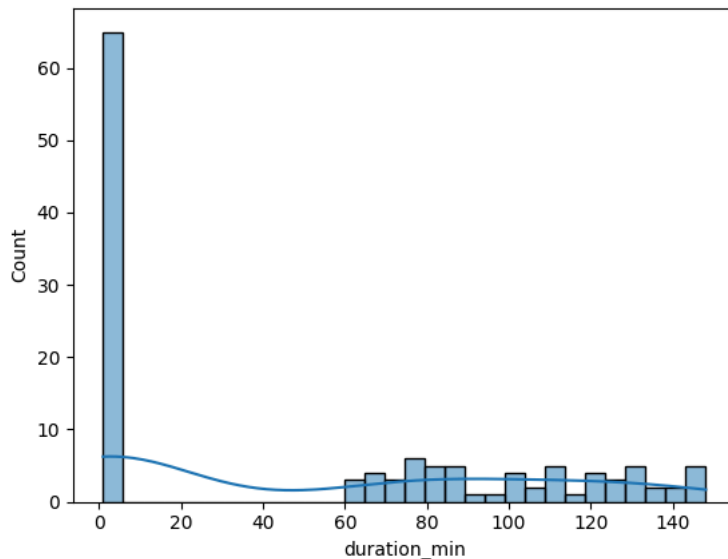
<Axes: xlabel='country'>



◆ Gemini

```
movie_df = df[df['type'] == 'Movie'].copy()
movie_df['duration_min'] = movie_df['duration'].str.extract('(\d+)').astype(float)
sns.histplot(movie_df['duration_min'], bins=30, kde=True)
```

```
<>:2: SyntaxWarning: invalid escape sequence '\d'
<>:2: SyntaxWarning: invalid escape sequence '\d'
/tmp/ipython-input-3901613910.py:2: SyntaxWarning: invalid escape sequence '\d'
  movie_df['duration_min'] = movie_df['duration'].str.extract('(\d+)').astype(float)
<Axes: xlabel='duration_min', ylabel='Count'>
```

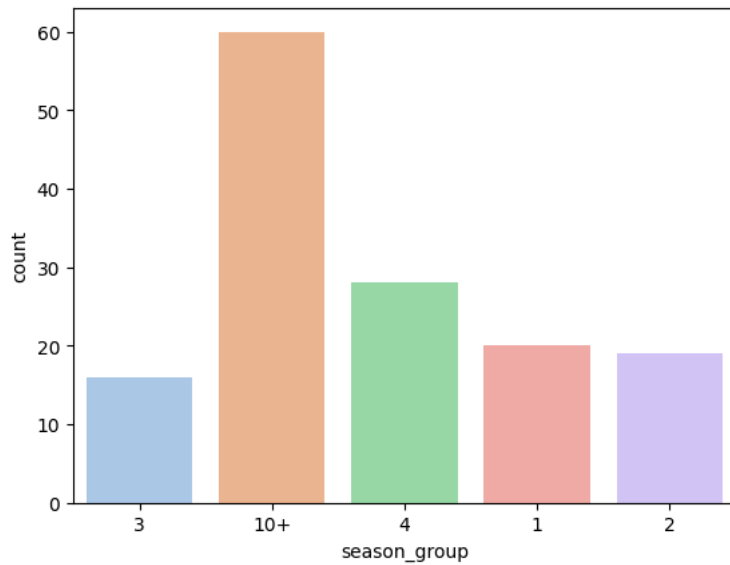


```
tv_df = df[df['type'] == 'TV Show'].copy()
tv_df['season_count'] = tv_df['duration'].str.extract('(\d+)')[0].fillna(0).astype(int)
tv_df['season_group'] = tv_df['season_count'].apply(lambda x: str(x) if x <= 10 else '10+')
sns.countplot(x='season_group', data=tv_df, palette='pastel')
```

```
<>:2: SyntaxWarning: invalid escape sequence '\d'
<>:2: SyntaxWarning: invalid escape sequence '\d'
/tmp/ipython-input-584744735.py:2: SyntaxWarning: invalid escape sequence '\d'
  tv_df['season_count'] = tv_df['duration'].str.extract('(\d+)')[0].fillna(0).astype(int)
/tmp/ipython-input-584744735.py:4: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and

```
sns.countplot(x='season_group', data=tv_df, palette='pastel')
<Axes: xlabel='season_group', ylabel='count'>
```



```
print("Final Cleaned Dataset Shape:", df.shape)
print("\n Project Completed Successfully")
```

Final Cleaned Dataset Shape: (269, 9)

Project Completed Successfully